which is a multiple of the object $0 \longrightarrow 0 \longrightarrow 1$. This results in a situation where the map A is injective and the map B is surjective (we rename the spaces to simplify notation):

$$\underset{V}{\bullet} \xhookrightarrow{A} \underset{W}{\bullet} \xrightarrow{B} \underset{Y}{\bullet}\,.$$

Next, let $X = \ker(B \circ A)$ and let X' be a complement of X in V. Let W' be a complement of $A(X)$ in W such that $A(X') \subset W'$. Then we get

$$\begin{aligned} &\underset{V}{\bullet} \xhookrightarrow{A} \underset{W}{\bullet} \xrightarrow{B} \underset{Y}{\bullet} \\ &\quad = \underset{X}{\bullet} \xrightarrow{A} \underset{A(X)}{\bullet} \xrightarrow{B} \underset{0}{\bullet} \;\oplus\; \underset{X'}{\bullet} \xhookrightarrow{A} \underset{W'}{\bullet} \xrightarrow{B} \underset{Y}{\bullet}\,. \end{aligned}$$

The first of these summands is a multiple of $1 \xrightarrow{\sim} 1 \longrightarrow 0$. Looking at the second summand, we now have a situation where A is injective, B is surjective, and furthermore $\ker(B \circ A) = 0$. To simplify notation, we redefine

$$V = X', \qquad W = W'.$$

Next we let $X = \operatorname{Im}(B \circ A)$ and let X' be a complement of X in Y. Furthermore, let $W' = B^{-1}(X')$. Then W' is a complement of $A(V)$ in W. This yields the decomposition

$$\begin{aligned} &\underset{V}{\bullet} \xhookrightarrow{A} \underset{W}{\bullet} \xrightarrow{B} \underset{Y}{\bullet} \\ &\quad = \underset{V}{\bullet} \xrightarrow[\sim]{A} \underset{A(V)}{\bullet} \xrightarrow[\sim]{B} \underset{X}{\bullet} \;\oplus\; \underset{0}{\bullet} \longrightarrow \underset{W'}{\bullet} \xrightarrow{B} \underset{X'}{\bullet}\,. \end{aligned}$$

Here, the first summand is a multiple of $1 \xrightarrow{\sim} 1 \xrightarrow{\sim} 1$. By splitting away the kernel of B, the second summand can be decomposed into multiples of $0 \longrightarrow 1 \xrightarrow{\sim} 1$ and $0 \longrightarrow 1 \longrightarrow 0$. So, on the whole, this quiver has six indecomposable representations:

$$1 \longrightarrow 0 \longrightarrow 0\,, \quad 0 \longrightarrow 0 \longrightarrow 1\,, \quad 1 \xrightarrow{\sim} 1 \longrightarrow 0\,,$$

$$1 \xrightarrow{\sim} 1 \xrightarrow{\sim} 1\,, \quad 0 \longrightarrow 1 \xrightarrow{\sim} 1\,, \quad 0 \longrightarrow 1 \longrightarrow 0\,.$$

(2) Now we look at the orientation

$$\bullet \longrightarrow \bullet \longleftarrow \bullet\,.$$